MATH 415 - FALL 2020 - EXTRA PRACTICE PROBLEMS - WEEK 6

1. The basics

1. *For each of the following lists of vectors, determine if they are linearly independent. If they are not linearly independent (and thus they are linearly dependent) give a* ***nontrivial*** *dependence relation for the vectors:*

(1) $\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}1\\1\\1\end{bmatrix}$

To determine if the three vectors are linearly independent, we need to see if the equation

$$x_1\begin{bmatrix}1\\0\\0\end{bmatrix} + x_2\begin{bmatrix}1\\1\\0\end{bmatrix} + x_3\begin{bmatrix}1\\1\\1\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$$

has a **nontrivial** solution. (Clearly $(x_1, x_2, x_3) = (0, 0, 0)$ will always be a solution, but we need to know if there is a second solution besides this one). Forming the augmented matrix

$$\left[\begin{array}{ccc|c}1&1&1&0\\0&1&1&0\\0&0&1&0\end{array}\right]$$

we see that the augmented matrix is already in Row Echelon Form. There are no free variables and the system is consistent. Therefore this is a unique solution, $(0, 0, 0)$. We conclude that the three original vectors are linearly independent.

(2) $\begin{bmatrix}1&1\\0&0\end{bmatrix}, \begin{bmatrix}-1&0\\0&0\end{bmatrix}, \begin{bmatrix}0&2\\0&0\end{bmatrix}$

For these three matrices, we have the following nontrivial linear combination:

$$\begin{bmatrix}1&1\\0&0\end{bmatrix} = -1\begin{bmatrix}-1&0\\0&0\end{bmatrix} + \frac{1}{2}\begin{bmatrix}0&2\\0&0\end{bmatrix}$$

2. *Find an example of*

(1) three distinct linearly dependent vectors in $\mathbb{R}^4$;

The vectors

$$\mathbf{x}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{x}_2 = \begin{pmatrix} 1 \\ -1 \\ 2 \\ 0 \end{pmatrix}, \quad \mathbf{x}_3 = \begin{pmatrix} 3 \\ 1 \\ 4 \\ 2 \end{pmatrix},$$

are linearly dependent because $2\mathbf{x}_1 + \mathbf{x}_2 - \mathbf{x}_3 = 0$. Another example is given by

$$\mathbf{x}_1 = \begin{pmatrix} 2 \\ -1 \\ 0 \\ 3 \end{pmatrix}, \quad \mathbf{x}_2 = \begin{pmatrix} 1 \\ 2 \\ 5 \\ -1 \end{pmatrix}, \quad \mathbf{x}_3 = \begin{pmatrix} 7 \\ -1 \\ 5 \\ 8 \end{pmatrix},$$

since $3\mathbf{x}_1 + \mathbf{x}_2 - \mathbf{x}_3 = 0$.

(2) three linearly independent vectors in $\mathbb{R}^6$;

The easiest case is given by

$$\mathbf{x}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{x}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{x}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

For the equation

$$c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + c_3\mathbf{x}_3 = 0$$

to hold, we need to have $c_1 = c_2 = c_3 = 0$, i.e. the trivial solution.

(3) three linearly dependent vectors in $\mathbb{R}^3$ such that each of these three vectors is not a scalar multiple of one of the other two vectors;

The vectors

$$\mathbf{x}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad \mathbf{x}_2 = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}, \quad \mathbf{x}_3 = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix},$$

are linearly dependent (check!). Clearly $\mathbf{x}_i \neq \alpha_i \mathbf{x}_j$ for $i \neq j$, i.e. $\mathbf{x}_1$ is not scalar multiple of $\mathbf{x}_2$ and not a scalar multiple of $\mathbf{x}_3$, and the same thing can be said of $\mathbf{x}_2$ and $\mathbf{x}_3$.

(4) three linearly dependent vectors in $\mathbb{R}^3$ such that any pair is of these three vectors is linearly independent.

Consider the vectors

$$\mathbf{x}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{x}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{x}_3 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$$

These vectors are linearly dependent in $\mathbb{R}^3$ since we have

$$c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + c_3\mathbf{x}_3 = 0$$

with $c_1 = c_2 = 1$ and $c_3 = -1$, i.e.

$$\mathbf{x}_1 + \mathbf{x}_2 - \mathbf{x}_3 = 0.$$

We now check the pair $\mathbf{x}_1$ and $\mathbf{x}_2$. We see that

$$c_1\mathbf{x}_1 + c_2\mathbf{x}_2 = 0 \Leftrightarrow c_1 = c_2 = 0$$

and hence $\mathbf{x}_1$ and $\mathbf{x}_2$ are linearly independent. Next,

$$c_1\mathbf{x}_1 + c_3\mathbf{x}_3 = 0 \Leftrightarrow c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + c_3 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = 0 \Leftrightarrow \begin{cases} c_1 + c_3 = 0 \\ c_3 = 0 \end{cases} \Leftrightarrow c_1 = c_3 = 0.$$

So that $\mathbf{x}_1$ and $\mathbf{x}_3$ are linearly independent. The case $\mathbf{x}_2$ and $\mathbf{x}_3$ works the same way.

3. *For each of the following matrices, find a minimal set of vectors* $\mathbf{a}_1, \ldots, \mathbf{a}_k$ *such that the column space of this matrix is equal to* $\text{span}\{\mathbf{a}_1, \ldots, \mathbf{a}_k\}$.

(1) $A = \begin{bmatrix} 1 & -2 & 1 \\ 2 & -4 & 1 \\ 1 & -2 & 1 \end{bmatrix}$

For A, $\text{Col}(A) = \text{Span}\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ -4 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \}$, however because the first two vectors are multiples of each other, we can remove one (it doesn't matter which) without changing the span. That is, $\text{Col}(A) = \text{Span}\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \}$. Moreover, $\text{Span}\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \}$ is a plane while $\text{Span}\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \}$ and $\text{Span}\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \}$ are distinct lines, so we cannot further remove vectors.

(2) $B = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

For B, $\text{Col}(B) = \text{span}\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \}$. It is easy to see that these three vectors are independent, thus we cannot remove any of them.

4. *Let* A *be an* 3×3*-matrix that has only one pivot position.*

(a) *Explain why there is a vector* $\mathbf{v} \in \mathbb{R}^3$ *such that* $\mathrm{Col}(A) = \mathrm{span}(\mathbf{v})$.

Let B be the row reduced echelon form of A. Let us write

$$A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 \end{bmatrix}, \qquad B = \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \mathbf{b}_3 \end{bmatrix}.$$

Since A has only one pivot position and so does B, we have $\mathrm{Col}(B) = \mathrm{span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$. In particular, there exists a column $\mathbf{b}_i = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$. For example, let us assume that $\mathbf{b}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, then we have $\mathbf{b}_2 = c\mathbf{b}_1$ and $\mathbf{b}_3 = c'\mathbf{b}_1$ for some scalars $c, c' \in \mathbb{R}$. By the definition of matrix multiplication, this is equivalent to

$$\begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \mathbf{b}_3 \end{bmatrix} \begin{bmatrix} c \\ -1 \\ 0 \end{bmatrix} = 0, \qquad \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \mathbf{b}_3 \end{bmatrix} \begin{bmatrix} c' \\ 0 \\ -1 \end{bmatrix} = 0.$$

Since A and B are row equivalent, the linear systems $Ax = 0$ and $Bx = 0$ have the same set of solutions. Therefore, we also have

$$\begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 \end{bmatrix} \begin{bmatrix} c \\ -1 \\ 0 \end{bmatrix} = 0, \qquad \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 \end{bmatrix} \begin{bmatrix} c' \\ 0 \\ -1 \end{bmatrix} = 0.$$

From here, we get $\mathbf{a}_2 = c\mathbf{a}_1$ and $\mathbf{a}_3 = c'\mathbf{a}_1$, and therefore $\mathrm{Col}(B) = \mathrm{span}\{\mathbf{a}_1\}$.

(b) *Is the same statement still true if* A *has two pivot positions?*

No. Consider

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

A is in the row reduced echelon form and clearly it has 2 pivots. However, the column space $\mathrm{Col}(A) = \mathrm{span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$, which cannot be written as a span of one single vector.

5. *Let* A *be an* $m \times n$*-matrix and* B *an* $n \times \ell$*-matrix. For each of the following statement, is it true or false? If true, explain why! If false, give a counterexample.*

(1) The column space of AB is contained in the column space of A.

True. The columns of AB are linear combinations of the columns of A. Thus every column of AB is in the column space of A, and hence every linear combination of the columns of AB is in the column space of A.

(2) The null space of AB contained in the null space A.

False. The null space of A is subspace of $\mathbb{R}^n$, while the null space of AB is a subspace of $\mathbb{R}^\ell$.

(3) The set of all vectors $b \in \mathbb{R}^m$ that are not in the column space of A, is a subspace of $\mathbb{R}^m$.

False. The zero vector is a linear combination of the columns of A. Thus the set described in the question does not contain the zero vector.

(4) If the column space of A only contains the zero vector, then A is the zero matrix.

True. If any of the columns of A would not be the zero vector, then this column would be in the column space of A.

(5) The column space of A is equal to the column space $2A$.

True. If $A = \begin{bmatrix}\mathbf{a}_1 & \dots & \mathbf{a}_n\end{bmatrix}$, then $2A = \begin{bmatrix}2\mathbf{a}_1 & \dots & 2\mathbf{a}_n\end{bmatrix}$. It is easy to check a vector is a linear combination of $\mathbf{a}_1, \dots, \mathbf{a}_n$ if and only if it is a linear of combination of $2\mathbf{a}_1, \dots, 2\mathbf{a}_n$.

(6) The column space of $A - I$ is equal to the column space of A.

False. Observe that the column space of $\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$ is not equal to the column space of $\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$.

6. *For each of the following statement, is it True or False? Give reasons!*

(1) The columns of a matrix A are linearly independent if the equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.

True. This follows from the definition of linearly independent.

(2) If S is a linearly dependent set, then each vector is a linear combination of the other vectors in S.

False. Consider the set
$$\left\{\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}2\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}\right\} \subseteq \mathbb{R}^2$$
which is linearly dependent. The third vector is not a linear combination of the first two vectors.

(3) The columns of any 4×5 matrix are linearly dependent.

True. The fact that there are more columns than rows guarantees free variables in the homogeneous equation $A\mathbf{x} = \mathbf{0}$. This will produce infinitely many dependence relations.

(4) If $\mathbf{x}$ and $\mathbf{y}$ are linearly independent, and if $\{\mathbf{x}, \mathbf{y}, \mathbf{z}\}$ is linearly dependent, then $\mathbf{z}$ is in $\text{span}\{\mathbf{x}, \mathbf{y}\}$.

True. Consider a nontrivial dependence relation $c_1\mathbf{x} + c_2\mathbf{y} + c_3\mathbf{z} = \mathbf{0}$. It must be the case that $c_3 \neq 0$, otherwise this will show that $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent. Thus $\mathbf{z} = -\frac{c_1}{c_3}\mathbf{x} - \frac{c_2}{c_3}\mathbf{y}$ and so $\mathbf{z} \in \text{span}\{\mathbf{x}, \mathbf{y}\}$.

(5) Two vectors are linearly dependent if and only if they lie on a line through the origin.

True. When **two** vectors are linearly dependent, it means that one vector is a scalar multiple of the other.

(6) If $\mathbf{x}$ and $\mathbf{y}$ are linearly independent, and if $\mathbf{z}$ is in $\text{span}\{\mathbf{x}, \mathbf{y}\}$, then $\{\mathbf{x}, \mathbf{y}, \mathbf{z}\}$ is linearly independent.

False. Just let $\mathbf{z} = \mathbf{x}$ and let $\mathbf{x}$ and $\mathbf{y}$ be any two linearly independent vectors.

(7) If a set in $\mathbb{R}^n$ is linearly dependent, then it contains at least $n+1$ vectors.

False. Consider

$$\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \end{bmatrix} \right\}$$

which is a linearly dependent set of vectors in $\mathbb{R}^2$ which contains only $2 < 2+1$ vectors.

(8) If $\mathbf{u}$ and $\mathbf{v}$ are linearly independent, then $\mathbf{u}+\mathbf{v}$ and $\mathbf{u}-\mathbf{v}$ are linearly independent.

True. We know that $\mathbf{u}$ and $\mathbf{v}$ are linearly independent. This means that the equation

$$a\mathbf{u} + b\mathbf{v} = 0$$

holds only for $a = b = 0$. Now $u+v$ and $u-v$ will be linearly independent if the only solution to

$$c(u+v) + d(u-v) = 0$$

is $c = d = 0$. Expanding, we see that

$$c\mathbf{u} + d\mathbf{u} + c\mathbf{v} - d\mathbf{v} = 0$$

so that

$$(c+d)\mathbf{u} + (c-d)\mathbf{v} = 0.$$

Solving, we see that this only happens when $c = d = 0$ since $\mathbf{u}$ and $\mathbf{v}$ are linearly independent. Hence $\mathbf{u}+\mathbf{v}$ and $\mathbf{u}-\mathbf{v}$ are linearly independent.

7. *Let $\mathbb{P}_2$ be the vector space of polynomials of degree at most 2.*

(1) Consider the following sets of polynomials in $\mathbb{P}_2$.

$$\mathcal{A} = \{1, t, t+1\}$$
$$\mathcal{B} = \{1, t, t^2\}$$
$$\mathcal{C} = \{1, 2t, t^2, t^2 - t\}.$$

Which of these sets are linearly independent?

$\mathcal{A}$ is obviously dependent, since $t+1 = 1+t$. $\mathcal{B}$ is the standard basis of $\mathbb{P}_2$, hence linearly independent. $\mathcal{C}$ is linearly dependent, since $t^2 - t = t^2 + \frac{1}{2}2t$.

(2) Now consider $\mathcal{D} = \{t^2+t+1, t+1, 1\}$. Is $\mathcal{D}$ linearly independent?

Suppose we have a linear combination

$$a(t^2+t+1) + b(t+1) + c = 0.$$

Our job is to show that $a = b = c = 0$ is the only solution. First its clear that $a = 0$ since the right hand side of the above expression is 0 so it has degree less than 2 as a polynomial. Thus the right hand side must have degree less than 2 also, so $a = 0$, for otherwise the right hand side would have degree 2, and t^2+t+1 is the only basis element that has a t^2 in it. Thus we have reduced to

$$b(t+1) + c = 0.$$

By a similar argument, it must also be the case that $b = 0$. But then we are left with $c = 0$.

(3) Write $q(t) = t^2 + 1$ as a linear combination of the elements of $\mathcal{D}$.

We have that
$$q(t) = t^2 + 1 = a(t^2 + t + 1) + b(t + 1) + c(1).$$
So we need to find a, b and c. This means that $a = 1$, $0 = a + b$ and $1 = a + b + c$, from which we get $a = c = 1$ and $b = -1$.

8. *Consider the vector space V of functions from $\mathbb{R}$ to $\mathbb{R}$. Are x, $\sin(x)$, $\cos(x)$ linear independent vectors in this vector space? If so, explain why.*

Yes, x, $\sin(x)$, $\cos(x)$ are linear independent vectors. Suppose that we have a linear combination
$$ax + b\sin(x) + c\cos(x) = 0.$$
Let $x = 0$. Then the left side becomes $a \cdot 0 + b \cdot 0 + c \cdot 1 = c$. Therefore by the assumption, we have $c = 0$. Similarly, if we let $x = \pi$, then we have $a \cdot \pi + b \cdot 0 + 0 = 0$, i.e. $a = 0$. Finally, since both a, c are zero, we must have $b\sin(x) = 0$, which implies $b = 0$.